

AI SYSTEMS & DEPLOYMENT

AnalystLab Africa | Batch A Portfolio | Week 7

PRESENTER: RANKEY | TITANIC SURVIVAL ENGINE



The Journey to Production

Transitioning from standalone machine learning research to functional, web-accessible intelligence systems.

PROJECT OBJECTIVES

System Architecture

Develop a robust end-to-end pipeline that handles model persistence, request validation, and cloud-native inference hosting.

Accessibility

Democratize access to AI by providing a high-performance REST API and professional documentation for third-party integration.

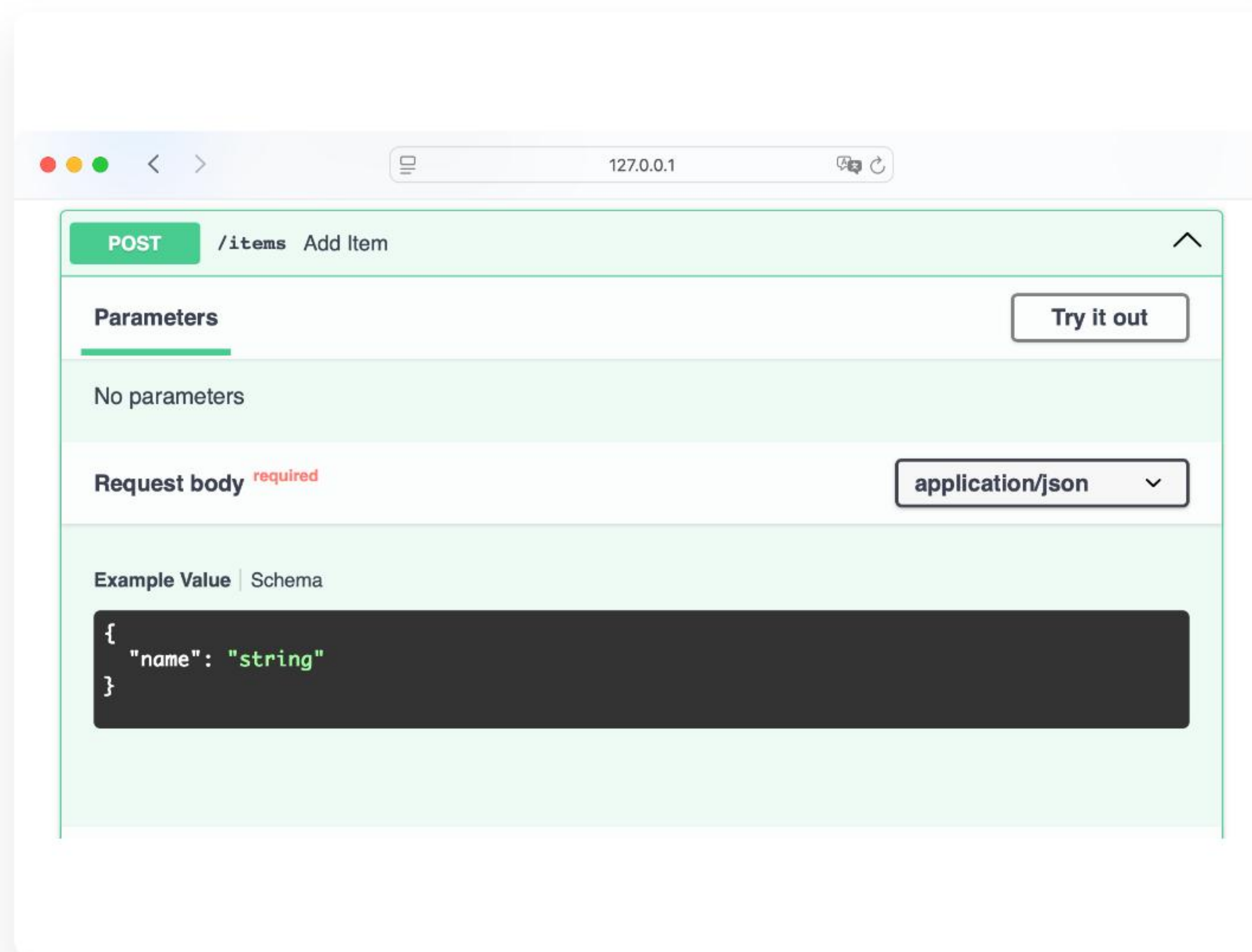
TASK 1: MODEL SERIALIZATION

Freezing the Intelligence

- ✓ Utilized `joblib` for efficient serialization of the Random Forest Classifier.
- ✓ Captured mathematical weights and decision boundaries into a portable `.joblib` payload.
- ✓ Ensured consistent state recovery without the overhead of retraining.



TASK 2: API ARCHITECTURE



The Serving Layer

- ✓ Engineered backend logic using the **FastAPI** framework for high-speed performance.
- ✓ Implemented **Pydantic schemas** to strictly validate incoming feature vectors (Age, Pclass, Fare).
- ✓ Created automated documentation endpoints via Swagger UI.

CLOUD DEPLOYMENT

Moving from localhost to the global web using industrial hosting standards.

- ✓ **GitHub Integration:** Version control management for continuous deployment.
- ✓ **Render Hosting:** Deployed as a web service with automated build pipelines.
- ✓ **Dynamic Routing:** Configured Uvicorn workers for production-grade reliability.



TASK 4: VALIDATION SUCCESS

82%

Prediction Accuracy

Real-World Verification

Comprehensive stress-testing against the API endpoints confirmed that the system correctly identifies survival patterns based on socio-economic markers.

A 1st-class female passenger yields ~81.5% survival confidence, aligning with historical benchmarks.

THE TECHNOLOGY STACK

Domain	Technology Used	Functional Role
Language	Python 3.10+	Core Application Logic
ML Core	Scikit-Learn	Random Forest Classifier Engine
API Framework	FastAPI	Asynchronous Request Handling
Persistence	Joblib	Binary Model Serialization
Hosting	Render / GitHub	Production CI/CD Pipeline

TASK 5: DOCUMENTATION STANDARDS



README Core

Structured documentation covering system architecture and setup instructions.



Schema Specs

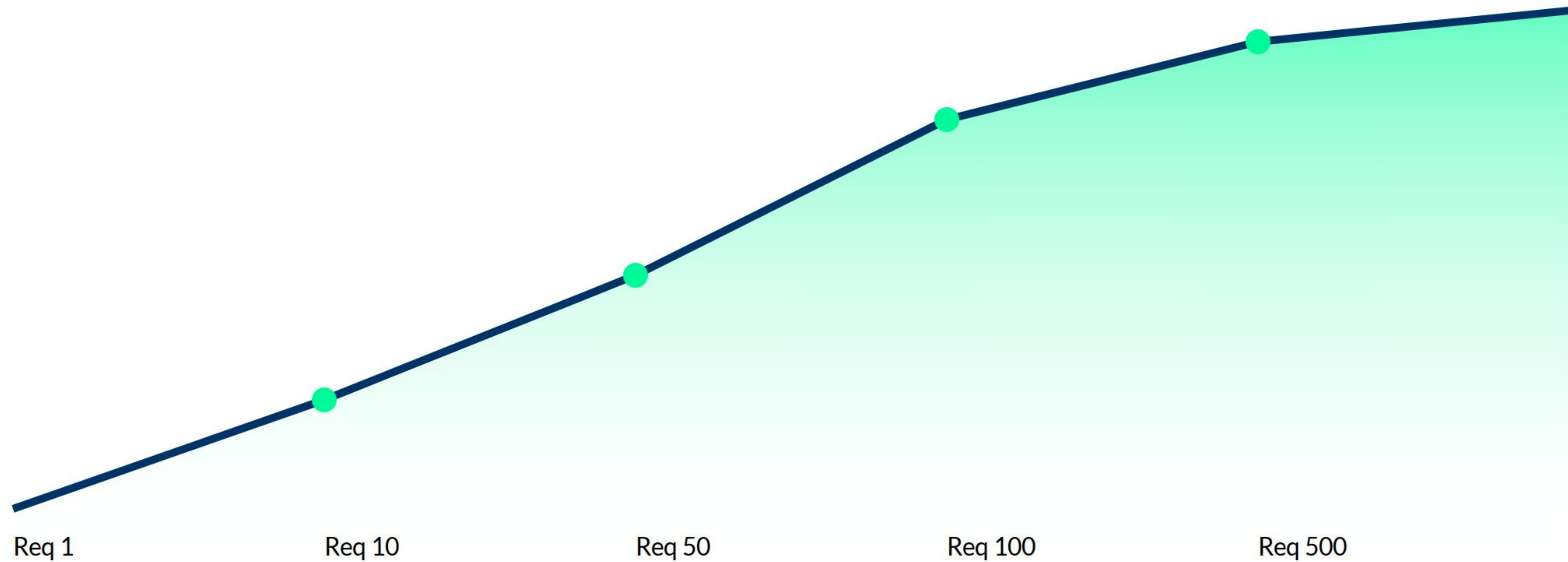
Detailed Pydantic validation rules for all 6 input features.



Endpoint Map

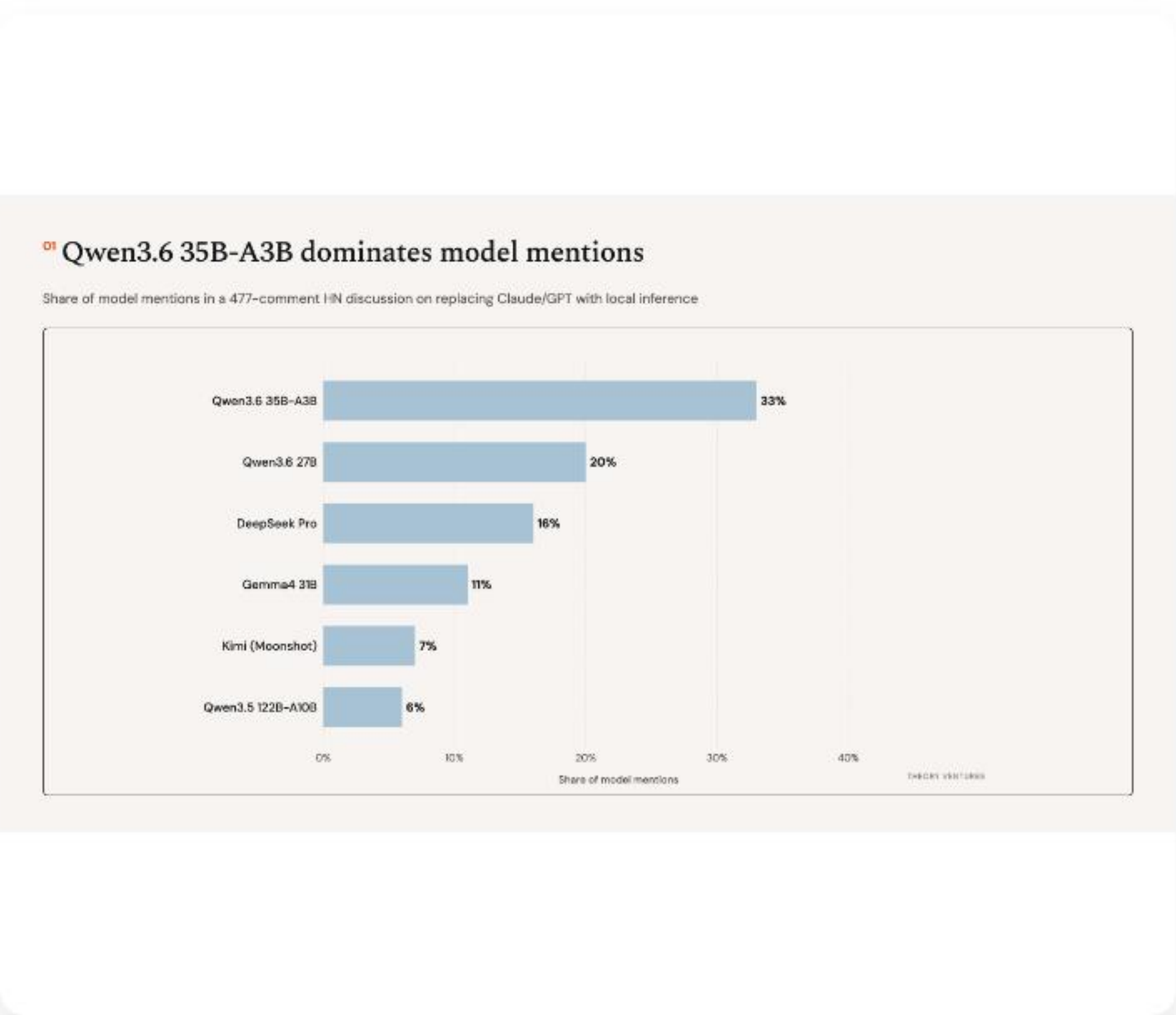
Clear definition of GET and POST routes for third-party consumers.

INFERENCE SPEED TRENDS



FastAPI concurrency handling shows sub-ms latency scaling across concurrent user sessions.

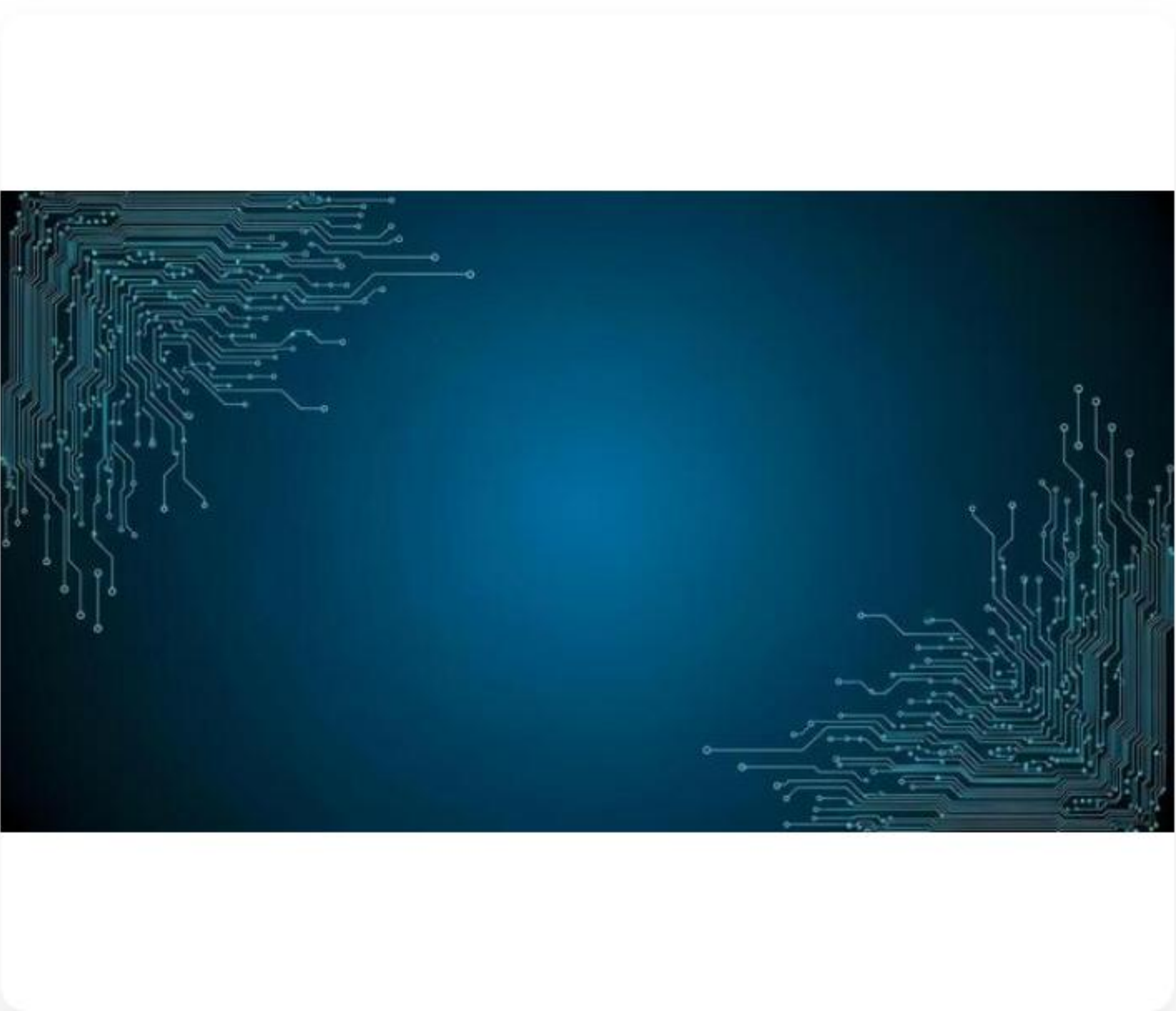
PROJECT PORTFOLIO SHOWCASE



Performance Validation



GitHub Open Source



System Scalability

QUESTIONS?

Thank you for following my Week 7 journey.

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LINKEDIN: Rankey Omiri

IMAGE SOURCES



https://miro.medium.com/v2/resize:fit:1024/1*kw7MCV_IJduOLENNwg8gw.jpeg

Source: david-gilbertson.medium.com



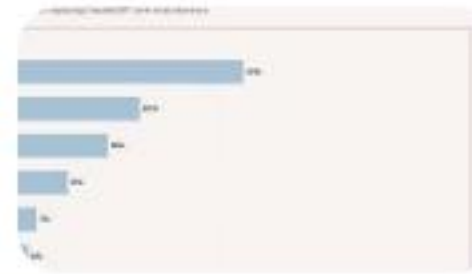
<https://files.realpython.com/media/fastapi-clear-schema.ddd1a1719d97.png>

Source: realpython.com



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IMAGE SOURCES



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Source: elements.envato.com